

PSEUDOCODE WRITING

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QUESTIONS

Q1

Student Management System

[BL3 – Apply]

10 marks

**Q2**

Library Book Management [BL3

– Apply]

10 marks

**Q3**

Employee Salary Calculation

[BL3 – Apply]

10 marks

**Q10**

10. Smart Home Device Controller [BL4 – Analyze]

10
marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock



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6. Bank Account Hierarchy [BL4

– Analyze]

10 marks

**Q7**7. Hospital Patient Record
System [BL4 – Analyze]

10 marks



Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

START

CLASS SmartDevice

METHOD executeCommand(cmd)

END CLASS

CLASS SmartLight EXTENDS SmartDevice

OVERRIDE executeCommand(cmd)

END CLASS

CLASS SmartThermostat EXTENDS SmartDevice

OVERRIDE executeCommand(cmd)

END CLASS

CLASS SmartLock EXTENDS SmartDevice

OVERRIDE executeCommand(cmd)

END CLASS

CREATE SmartLight l

CREATE SmartThermostat t

CREATE SmartLock s

CALL l.executeCommand("ON")

CALL t.executeCommand("STATUS")

CALL s.executeCommand("OFF")

STOP

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QUESTIONS

Q1

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[BL3 – Apply]

10 marks

Q2

Library Book Management [BL3

– Apply]

10 marks

Q3

Employee Salary Calculation

[BL3 – Apply]

10 marks

Q9

9. University Staff Hierarchy [BL4 – Analyze]

10

marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure. Justify the use of multilevel



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Q6

6. Bank Account Hierarchy [BL4

– Analyze]

10 marks

Q7

7. Hospital Patient Record

System [BL4 – Analyze]

10 marks

Q88. E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**9. University Staff Hierarchy
[BL4 – Analyze]

10 marks

**Q10**10. Smart Home Device
Controller [BL4 – Analyze]

10 marks



10/10 answered

```
    OVERRIDE calculateAllowance()  
END CLASS
```

```
CLASS Professor EXTENDS TeachingStaff  
    OVERRIDE calculateAllowance()  
END CLASS
```

```
CLASS Lecturer EXTENDS TeachingStaff  
    OVERRIDE calculateAllowance()  
END CLASS
```

```
CLASS NonTeachingStaff EXTENDS Staff  
    OVERRIDE calculateAllowance()  
END CLASS
```

```
CREATE Professor p  
CREATE Lecturer l  
CREATE NonTeachingStaff n
```

```
CALL p.calculateAllowance()  
CALL l.calculateAllowance()  
CALL n.calculateAllowance()
```

```
STOP
```

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10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks



Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks



Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks



Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks



Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks



• Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts:
Polymorphism | Inheritance | Method Overriding

Your Answer

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

START

CLASS Patient

patientId, name

METHOD generateBill()

END CLASS

CLASS InPatient EXTENDS Patient

wardNo, dailyCharges

OVERRIDE generateBill()

END CLASS

CLASS OutPatient EXTENDS Patient

appointmentDate, consultationFee

OVERRIDE generateBill()

END CLASS

CREATE InPatient i

CREATE OutPatient o

CALL i.generateBill()

CALL o.generateBill()

STOP

 Save Answer

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10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks



Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks



Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks



Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks



Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks



• OutPatient — with appointment date and
consultation fee

Analyze encapsulation requirements for sensitive
data (diagnosis, prescription). Override a
generateBill() method for each type and compare the
billing logic differences. OOP Concepts:
Encapsulation | Inheritance | Polymorphism

Your Answer

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

START

CLASS Product

METHOD applyDiscount()

END CLASS

CLASS Electronics EXTENDS Product

OVERRIDE applyDiscount()

END CLASS

CLASS Clothing EXTENDS Product

OVERRIDE applyDiscount()

END CLASS

CLASS Grocery EXTENDS Product

OVERRIDE applyDiscount()

END CLASS

CREATE Electronics e

CREATE Clothing c

CREATE Grocery g

CALL e.applyDiscount()

CALL c.applyDiscount()

CALL g.applyDiscount()

STOP



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10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]

10 marks



Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]

10 marks



Q5

5. Vehicle Registration System
[BL3 – Apply]

10 marks



Q6

6. Bank Account Hierarchy [BL4
– Analyze]

10 marks



Q7

7. Hospital Patient Record
System [BL4 – Analyze]

10 marks



- Inheritance of common attributes (accno, balance, holder)

- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

START

CLASS BankAccount

accNo, balance, holder

METHOD withdraw()

END CLASS

CLASS SavingsAccount EXTENDS BankAccount

METHOD calculateInterest()

OVERRIDE withdraw()

END CLASS

CLASS CurrentAccount EXTENDS BankAccount

METHOD overdraft()

OVERRIDE withdraw()

END CLASS

CREATE SavingsAccount s

CREATE CurrentAccount c

CALL s.withdraw()

CALL c.withdraw()

STOP

 Save Answer



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**Q4**

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

**Q5**

5. Vehicle Registration System
[BL3 – Apply]
10 marks

**Q6**

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

**Q7**

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

**Q5**

5. Vehicle Registration System [BL3 – Apply]

10
marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

START

CLASS Vehicle

regNo, owner, fuelType

METHOD displayDetails()

END CLASS

CLASS TwoWheeler EXTENDS Vehicle

engineCapacity

OVERRIDE displayDetails()

END CLASS

CLASS FourWheeler EXTENDS Vehicle

seatingCapacity, ACstatus

OVERRIDE displayDetails()

END CLASS

CREATE TwoWheeler t

CREATE FourWheeler f

CALL t.displayDetails()

CALL f.displayDetails()

STOP

 Save Answer



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QUESTIONS

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10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q4

Shape Area Calculator using Polymorphism [BL3 – Apply]

10
marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

```
CLASS Circle EXTENDS Shape
    area = 3.14 * radius * radius
END CLASS
```

```
CLASS Rectangle EXTENDS Shape
    area = length * breadth
END CLASS
```

```
CLASS Triangle EXTENDS ShapeP
    area = (base * height) / 2
END CLASS
```

```
CREATE Circle c
CREATE Rectangle r
CREATE Triangle t
```

```
CALL c.calculateArea()
CALL r.calculateArea()
CALL t.calculateArea()
```

```
STOP
```

 Save Answer



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[BL3 – Apply]

10 marks

**Q4**

Shape Area Calculator using

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10 marks

**Q5**

5. Vehicle Registration System

[BL3 – Apply]

10 marks

**Q6**

6. Bank Account Hierarchy [BL4

– Analyze]

10 marks

**Q7**

7. Hospital Patient Record

System [BL4 – Analyze]

10 marks

**Q3**

Employee Salary Calculation [BL3 – Apply]

10
marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to:

- Calculate gross salary (basic + HRA + DA)

- Calculate net salary after deductions (PF, tax)

- Display pay slip

Apply data hiding so salary components are not directly accessible.

OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

START

CLASS Employee

PRIVATE: empld, name, basicPay, designation

METHOD acceptDetails()

INPUT empld, name, basicPay, designation

END METHOD

METHOD calculateSalary()

$$\text{gross} = \text{basicPay} + (\text{basicPay} * 20/100) +$$
$$(\text{basicPay} * 10/100)$$
$$\text{net} = \text{gross} - (\text{basicPay} * 5/100) - (\text{basicPay} * 2/100)$$

END METHOD

METHOD displayPaySlip()

PRINT empld, name, gross, net

END METHOD

END CLASS

CREATE Employee e

CALL e.acceptDetails()

CALL e.calculateSalary()

CALL e.displayPaySlip()

STOP

 Save Answer



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10 marks

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Library Book Management [BL3

– Apply]

10 marks

**Q3**

Employee Salary Calculation

[BL3 – Apply]

10 marks

**Q4**

Shape Area Calculator using

Polymorphism [BL3 – Apply]

10 marks

**Q5**

5. Vehicle Registration System

[BL3 – Apply]

10 marks

**Q6**

6. Bank Account Hierarchy [BL4

– Analyze]

10 marks

**Q7**

7. Hospital Patient Record

System [BL4 – Analyze]

10 marks

**Q2**

Library Book Management [BL3 – Apply]

10

marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

**Q10**

10. Smart Home Device
Controller [BL4 – Analyze]
10 marks



10/10 answered

START

CLASS Book

PRIVATE

title
author
ISBN
isAvailable

PUBLIC

METHOD setBook(t, a, i)

title ← t
author ← a
ISBN ← i
isAvailable ← TRUE

END METHOD

METHOD issueBook()

IF isAvailable = TRUE THEN
isAvailable ← FALSE
PRINT "Book Issued"
ELSE
PRINT "Book Not Available"
END IF

END METHOD

METHOD returnBook()

isAvailable ← TRUE

 Save Answer



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QUESTIONS

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10 marks

**Q3**

Employee Salary Calculation
[BL3 – Apply]
10 marks

**Q4**

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

**Q5**

5. Vehicle Registration System
[BL3 – Apply]
10 marks

**Q6**

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

**Q7**

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

**Q1**

Student Management System [BL3 – Apply]

10
marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

Q8

8. E-Commerce Product Catalog

[BL4 – Analyze]

10 marks

**Q9**

9. University Staff Hierarchy

[BL4 – Analyze]

10 marks

**Q10**10. Smart Home Device
Controller [BL4 – Analyze]

10 marks



10/10 answered

START

CLASS Student

PRIVATE

name

rollNo

marks1, marks2, marks3

average

PUBLIC

METHOD acceptDetails()

INPUT name

INPUT rollNo

INPUT marks1

INPUT marks2

INPUT marks3

END METHOD

METHOD calculateAverage()

average ← (marks1 + marks2 + marks3) / 3

END METHOD

METHOD displayResult()

PRINT "Name: ", name

PRINT "Roll No: ", rollNo

PRINT "Average: ", average

IF average >= 50 THEN

PRINT "Result: PASS"

ELSE

PRINT "Result: FAIL"

END IF

END METHOD

END CLASS

MAIN

CREATE object s OF Student

CALL s.acceptDetails()

 Save Answer

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